

Probabilistic Machine Learning

Course Subject/Number BST570-1 (4 credits)

Term and Year Fall 2026

Day/Time and Location MW 1:30–2:40 PM; Th 1:30–2:20 PM. Saunders Research Building 1.404.

Instructor Seong-Hwan Jun

Office Hours By appointment.

Contact Information

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Academic Integrity: You may discuss homework problems with others, but you must not retain written notes from your conversations with other students, or share data via computer files to be used in completing your homework. Your written work must be completed without reference to such notes, with the exception of class and recitation notes, which may be retained in written form.

Use of Artificial Intelligence for homeworks and labs: You may discuss homework problems with an AI chatbot, but you may not submit AI-generated code verbatim. Attempt each problem on your own first, and only then use AI to help you modify and adapt your solution. Your written explanations must be your own, although using AI to improve clarity or correct grammar is acceptable. Homeworks will be discussed in class, so it is your responsibility to understand what you submitted and to be able to explain it. Use of AI for the labs is discouraged. The labs are graded for completeness rather than correctness, and reading the documentation for the functions and packages you use is part of the exercise.

Disability resources: The University of Rochester respects and welcomes students of all backgrounds and abilities. In the event you encounter any barrier(s) to full participation in this course due to the impact of a disability, please contact the Office of Disability Resources. The access coordinators in the Office of Disability Resources can meet with you to discuss the barriers you are experiencing and explain the eligibility process for establishing academic accommodations. You can reach the Office of Disability Resources at: disability@rochester.edu; (585) 276-5075; Taylor Hall; www.rochester.edu/college/disability.

Course Description This course provides an introduction to probabilistic modeling techniques and inference methods. Topics covered include graphical models and inference methods. Strong emphasis is placed on implementation using Python, utilizing libraries such as PyTorch. The students will gain hands-on experiences developing probabilistic models and inference methods for various applications.

Course Objectives

- Understand the foundational concepts of probabilistic reasoning.

- Learn to formulate and solve inference problems using probabilistic models.
- Gain practical experience with probabilistic machine learning tools and libraries.

Assessments

- Assignments (5–6 over the term): 40%.
- 2 Quizzes: 20%.
- Labs, graded for completion and participation: 10%.
- Paper presentation: 30%.

Total: 100%.

Submission of labs and assignments: Each student receives a private GitHub repository for the course. For each lab or assignment, create a branch, commit your Quarto (`.qmd`) file to it, push, and open a pull request into `main` before the due date; the time the pull request is opened is the submission time. Feedback is given as comments on the pull request. Grades are recorded in Blackboard. Setting up the repository is part of Lab 1.

Grading Procedures

- A: 90-100%
- A-: 85-90%
- B+: 80-85%
- B: 75-79%
- B-: 70-74%
- C: 60-69%
- D: 50-59%
- E: < 50%

Textbooks

- *Machine Learning: a Probabilistic Perspective* by Kevin P. Murphy or
- *Pattern Recognition and Machine Learning* by Christopher Bishop. PDF available online: <https://www.microsoft.com/en-us/research/people/cmbishop/prml-book/>.
- *Probabilistic Machine Learning: An Introduction* by Kevin P. Murphy. <https://probml.github.io/pml-book/book1.html>.
- *Probabilistic Machine Learning: Advanced Topics* by Kevin Murphy. Draft PDF available online: <https://probml.github.io/pml-book/book2.html>.

Course Outline

Background

- Probability theory and random variables.
- Probabilistic reasoning and Bayesian statistics.
- Graphical models.
- Optimization and sampling.
- Intro to Python programming and Python libraries for machine learning.

Exact Inference

- Directed and acyclic graphical models: Bayesian networks, trees, and hidden Markov models.
- Belief propagation and variable elimination.
- Factor graphs and message passing.

Asymptotically Exact Inference

- Directed and undirected graphical models: Markov random fields, state space models, mixture models
- Markov chain Monte Carlo methods (Metropolis-Hastings, Gibbs, and gradient-based methods).
- Sequential Monte Carlo methods.

Approximate Inference

- Kullback-Leibler divergence and evidence lower bound optimization (ELBO).
- Mean-field variational inference and coordinate ascent VI with applications to Gaussian mixture model and latent Dirichlet Allocation.
- Variational auto-encoders.
- Probabilistic programming languages for blackbox variational inference.
- Topics: e.g., normalizing flows, expectation propagation, transformers.